Lec. 4. Applications of Quantum Theory to Simple Systems

4.1. Introduction

The three basic modes of motion (translation, vibration, and rotation) all play an important role in chemistry, because those are the ways in which molecules store energy. For example, gaseous molecules undergo translational motion and their kinetic energy is a contribution to the total internal energy. Molecules can also store energy as rotational kinetic energy and transitions between their rotational energy states can be observed spectroscopically. Energy is also stored as molecular vibration, and transitions between vibrational states are responsible for the appearance of infrared and Raman spectra.

At this point, we will limit our discussion to translational motion, and in section 3.1, we learned that the Schrödinger equation for a particle moving freely in one dimension (V = 0), is as follows:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \qquad (4.1)$$

or more succinctly

$$\hat{H}\psi = E\psi \qquad \hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} \qquad (4.2)$$

The general solutions of the equation 4.1 are as follows:

$$\psi_k = Ae^{ikx} + Be^{-ikx} \qquad E_k = \frac{k^2\hbar^2}{2m} \qquad (4.3)$$

Here, it has to be noted that both the wavefunctions and the energies (that is, the eigenfunctions and eigenvalues of $\hat{H}$), have been labeled with the index k. We can verify that these functions are solutions, by substituting ψ_k into the left-hand side of equation 4.1 and showing that the result is equal to $E_k\psi_k$. In this case, all values of k, and therefore all values of the energy, are permitted. It means that the translational energy of a free particle is not quantized.

In section 3.5, we learned that a wavefunction of the form e^{ikx} describes a particle with linear momentum $p_x = +k\hbar$, corresponding to motion towards positive x (to the right), and that a wavefunction of the form e^{-ikx} describes a particle with the same magnitude of linear momentum but travelling towards negative x (to the left). That is, e^{ikx} is an eigenfunction of the operator $\hat{p}_x$ with eigenvalue $+k\hbar$ and e^{-ikx} is an eigenfunction with eigenvalue $-k\hbar$. In both states, $|\psi|^2$ is independent of x, which implies that the position of the particle is completely unpredictable.

4.2. A Particle in a Box

Let us consider a particle in a box, in which a particle of mass m is confined between two walls at $x = 0$ and $x = L$. The potential energy is zero inside the box but rises suddenly to infinity at the walls (Fig. 4.1). This model is an idealization of the potential energy of a gaseous molecule that is free to move in a one-dimensional container. The particle in a box is also used in statistical thermodynamics in assessing the contribution of the translational motion of molecules to their thermodynamic properties.

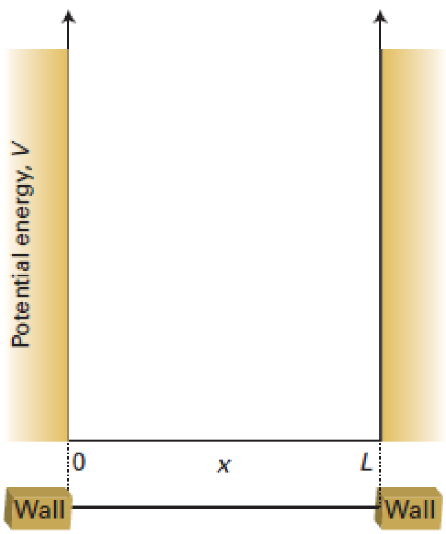

Fig. 4.1. A Particle in a Box

The Schrödinger equation for the region between the walls (where $V = 0$) is the same as for a free particle (equation 4.1), so the general solutions given in equation 4.3 are also the same.

However, it will be an advantage to use the relation $e^{\pm ix} = \cos x \pm i \sin x$, to write

$\psi_k = Ae^{ikx} + Be^{-ikx} = A(\cos kx + i \sin kx) + B(\cos kx - i \sin kx)$

$\quad = (A+B)\cos kx + (A - B)\, i \sin kx$

If we absorb all numerical factors into two new coefficients C and D, then the general solution will take the following form.

$$\psi_k(x) = C \sin kx + D \cos kx \qquad E_k = \frac{k^2\hbar^2}{2m} \tag{4.4}$$

For a free particle, any value of E_k corresponds to an acceptable solution. However, when the particle is confined within a region, the acceptable wavefunctions must satisfy certain boundary conditions, or constraints on the function at certain locations. Further, it is physically impossible for the particle to be found with an infinite potential energy. We can say that the wavefunction must be zero where V is infinite, at $x < 0$ and $x > L$. It means that the continuity of the wavefunction requires it to disappear at $x = 0$ and $x = L$.

So, the boundary conditions are $\psi_k(0) = 0$ and $\psi_k(L) = 0$.

Let us consider the wall at $x = 0$.

According to equation 4.4, $\psi(0) = D$ (since $\sin 0 = 0$ and $\cos 0 = 1$).

However, because $\psi(0) = 0$; $D = 0$.

It follows that the wavefunction must be of the form $\psi_k(x) = C \sin kx$.

The value of ψ at the other wall (at $x = L$) is $\psi_k(L) = C \sin kL$, which must also be zero.

Taking $C = 0$ would give $\psi_k(x) = 0$ for all x, which would conflict with the Born interpretation (the particle must be somewhere). As such, kL need to be selected so that $\sin kL \neq 0$, which is satisfied by

$kL = n\pi; \qquad n = 1, 2, 3, \ldots..$

The value $n = 0$ is ruled out, because it implies $k = 0$ and $\psi_k(x) = 0$ everywhere (because $\sin 0 = 0$), which is unacceptable.

Negative values of n merely change the sign of $\sin kL$ (because $\sin(-x) = -\sin x$) and do not give rise to a new wavefunction.

The wavefunctions are therefore

$$\psi_n(x) = C \sin(n\pi x/L); \qquad n = 1, 2, 3, \ldots.. \qquad (4.5)$$

Here, we have labeled the solutions with the index n instead of k.

Since $E_k = k^2\hbar^2/2m$ and $k = n\pi/L$, it follows that the energy of the particle is limited to the values $n^2h^2/8mL$, with n = 1, 2, 3, …..

Thus, we can conclude that the energy of the particle in a one-dimensional box is quantized and that this quantization arises from the boundary conditions that ψ must satisfy if it is to be an acceptable wavefunction.

This is a general conclusion: "the need to satisfy boundary conditions implies that only certain wavefunctions are acceptable, and hence restricts observables to discrete values". So far, we have seen that energy has been quantized; but it has to be noted that other physical observables may also be quantized.

We can complete the derivation of the wavefunctions by finding the normalization constant. To do so, we will have to look for the value of C that ensures that the integral of ψ^2 over all the space available to the particle (i.e. from $x = 0$ to $x = L$) is equal to 1.

$$\int_0^L \psi^2 \, dx = C^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = C^2 \times \frac{L}{2} = 1 \qquad (4.6)$$

It follows that $C = (2/L)^{1/2}$.

Therefore, the complete solution to the problem is

$$E_n = \frac{n^2h^2}{8mL^2} \qquad n = 1, 2, \ldots \qquad (4.7)$$

$$\psi_n(x) = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n\pi x}{L}\right) \qquad \text{for } 0 \le x \le L \tag{4.8}$$

The energies and wavefunctions are labeled with the 'quantum number' n. A quantum number is an integer that labels the state of the system. For a particle in a box there are an infinite number of acceptable solutions, and the quantum number n specifies the one of interest (Fig. 4.2). As well as acting as a label, a quantum number can be used to calculate the energy corresponding to the state and to write down the wavefunction explicitly (in the present example, by using equations 4.7 and 4.8).

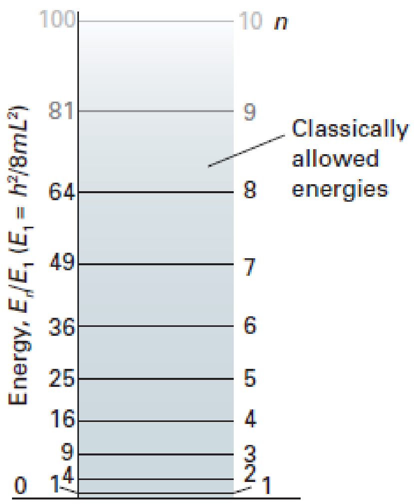

Fig. 4.2. Allowed Energy Levels for a Particle in a Box

The linear momentum of a particle in a box is not well-defined because the wavefunction sin kx is not an eigenfunction (like cos kx) of the linear momentum operator. However, each wavefunction is a superposition of momentum eigenfunctions.

$$\psi_n = \left(\frac{2}{L}\right)^{1/2} \sin\frac{n\pi x}{L} = \frac{1}{2\mathrm{i}}\left(\frac{2}{L}\right)^{1/2}(\mathrm{e}^{\mathrm{i}kx} - \mathrm{e}^{-\mathrm{i}kx}) \qquad k = \frac{n\pi}{L} \tag{4.9}$$

It follows that measurement of the linear momentum will give the value $+k\hbar$ for half the measurements of momentum and $-k\hbar$ for the other half. This detection of opposite directions of travel with equal probability is the quantum mechanical version of the classical picture that a particle in a box moves from wall to wall and in any given period spends half its time travelling to the left and half travelling to the right.

Because n cannot be zero, the lowest energy that the particle may possess is not zero (as it would be allowed by classical mechanics, corresponding to a stationary particle), and will be as follows.

$$E_1 = \frac{h^2}{8mL^2} \qquad (4.10)$$

This lowest, irremovable energy is called the zero-point energy. The physical origin of the zero-point energy can be explained in two ways. First, the uncertainty principle requires a particle to possess kinetic energy if it is confined to a finite region: the location of the particle is not completely indefinite, so its momentum cannot be precisely zero. Hence, it has nonzero kinetic energy. Second, if the wavefunction is to be zero at the walls, but smooth, continuous, and not zero everywhere, then it must be curved, and curvature in a wavefunction implies the possession of kinetic energy.

The separation between adjacent energy levels with quantum numbers n and $n + 1$ is

$$E_{n+1} - E_n = \frac{(n+1)^2 h^2}{8mL^2} - \frac{n^2 h^2}{8mL^2} = (2n+1)\frac{h^2}{8mL^2} \qquad (4.11)$$

This separation decreases as the length of the container increases, and is very small when the container has macroscopic dimensions. The separation of adjacent levels becomes zero when the walls are infinitely far apart. The translational energy of completely free particles (those not confined by walls) is not quantized.

The probability density for a particle in a box is

$$\psi^2(x) = \frac{2}{L}\sin^2\frac{n\pi x}{L} \qquad (4.12),$$

and varies with position.

The non-uniformity is pronounced when n is small, but $\psi^2(x)$ becomes more uniform as n increases. The distribution at high quantum numbers reflects the classical result that a particle bouncing between the walls spends, on the average, equal times at all points. The fact that the quantum result corresponds to the classical prediction at high quantum numbers is an illustration of the correspondence principle, which states that classical mechanics emerges from quantum mechanics as high quantum numbers are reached.

Question: What is the probability of locating a particle between $x = 0$ (the left-hand end of a box) and $x = 0.2$ nm in its lowest energy state in a box of length 1.0 nm?

Method: The value of $\psi^2 dx$ is the probability of finding the particle in the small region dx located at x. So, the total probability of finding the particle in the specified region is the integral of $\psi^2 dx$ over that region. The wavefunction of the particle is given in equation 4.8 with $n = 1$.

Answer: The probability of finding the particle in a region between $x = 0$ and $x = l$ is

$$P = \int_0^l \psi_n^2\, dx = \frac{2}{L}\int_0^l \sin^2\frac{n\pi x}{L}\, dx = \frac{l}{L} - \frac{1}{2n\pi}\sin\frac{2\pi n l}{L}$$

Since $n = 1$ and $l = 0.2$ nm, $P = 0.05$.

The result corresponds to a chance of 1 in 20 of finding the particle in the region. As n becomes infinite, the sine term, which is multiplied by $1/n$, makes no contribution to P and the classical result, $P = 1/L$, is obtained.

4.3. Motion in Two and Three Dimensions

4.3.1. Two Dimensions

Next, let us consider a two-dimensional version of the particle in a box. Now the particle is confined to a rectangular surface of length L_1 in the x-direction and L_2 in the y-direction. The potential energy is zero everywhere except at the walls, where it is infinite (Fig. 4.3).

The wavefunction is now a function of both x and y, and the Schrödinger equation is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}\right) = E\psi \qquad (4.13)$$

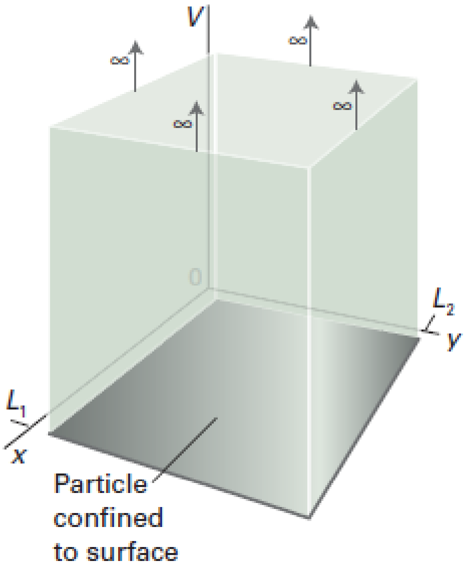

Fig. 4.3. A Two-Dimensional Square Well

Now, we have to solve this partial differential equation, a differential equation in more than one variable. Some partial differential equations can be simplified by the separation of variables technique, which divides the equation into two or more ordinary differential equations, one for each variable.

An important application of this procedure is the separation of the Schrödinger equation for the hydrogen atom into equations that describe the radial and angular variation of the wavefunction. This technique is particularly simple for a two-dimensional square well, as can be seen by testing whether a solution of equation 4.13 can be found by writing the wavefunction as a product of functions, one depending only on x and the other only on y.

$$\psi(x, y) = X(x)\,Y(y) \qquad (4.14)$$

With this substitution, we can separate equation 4.13 into two ordinary differential equations, one for each coordinate:

$$-\frac{\hbar^2}{2m}\frac{d^2X}{dx^2} = E_X X \qquad -\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2} = E_Y Y \qquad E = E_X + E_Y \qquad (4.15)$$

The quantity E_X is the energy associated with the motion of the particle parallel to the x-axis, and E_Y is the energy associated with the motion parallel to the y-axis.

Similarly, $X(x)$ is the wavefunction associated with the particle's freedom to move parallel to the x-axis and $Y(y)$ is the wavefunction associated with the motion parallel to the y-axis.

Each of the two ordinary differential equations in equation 4.15 is the same as the one dimensional square-well Schrödinger equation. As such, we can adapt the results in equation 4.8 without further calculations.

$$X_{n_1}(x) = \left(\frac{2}{L_1}\right)^{1/2} \sin\frac{n_1 \pi x}{L_1} \qquad Y_{n_2}(y) = \left(\frac{2}{L_2}\right)^{1/2} \sin\frac{n_2 \pi y}{L_2} \tag{4.16}$$

Then, because $\psi = XY$ and $E = E_X + E_Y$, we get the wavefunctions and energies of a particle in two-dimensional box as follows.

$$\psi_{n_1,n_2}(x,y) = \frac{2}{(L_1 L_2)^{1/2}} \sin\frac{n_1 \pi x}{L_1} \sin\frac{n_2 \pi y}{L_2} \tag{4.17}$$

$$E_{n_1,n_2} = \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2}\right)\frac{h^2}{8m} \qquad 0 \le x \le L_1, 0 \le y \le L_2 \tag{4.18}$$

with the quantum numbers taking the values n_1 = 1,2,3, ….. and n_2 = 1,2,3, ….. independently.

4.3.2. Three Dimensions

We treat a particle in a three-dimensional box in the same way. The wavefunctions have another factor (for the z-dependence), and the energy has an additional term in n_3^2/L_3^2.

Then, the solution of the Schrödinger equation by the separation of variables technique gives

$$\psi_{n_1,n_2,n_3}(x,y,z) = \left(\frac{8}{L_1 L_2 L_3}\right)^{1/2} \sin\frac{n_1 \pi x}{L_1} \sin\frac{n_2 \pi y}{L_2} \sin\frac{n_3 \pi z}{L_3} \tag{4.19}$$

$$E_{n_1,n_2,n_3} = \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2}\right)\frac{h^2}{8m} \qquad 0 \le x \le L_1, 0 \le y \le L_2, 0 \le z \le L_3 \tag{4.20}$$

with the quantum numbers taking the values n_1 = 1,2,3, ….., n_2 = 1,2,3, ….. and n_3 = 1,2,3, ….. independently.

4.3.3. Degeneracy

An interesting feature of the solutions for a particle in a two-dimensional box is obtained when the plane surface is square, with $L_1 = L_2 = L$.

Then, the equations 4.17 and 4.18 become

$$\psi_{n_1,n_2}(x,y) = \frac{2}{L}\sin\frac{n_1\pi x}{L}\sin\frac{n_2\pi y}{L} \qquad E_{n_1,n_2} = (n_1^2 + n_2^2)\frac{h^2}{8mL^2} \tag{4.21}$$

Considering the cases $n_1 = 1$, $n_2 = 2$ and $n_1 = 2$ and $n_2 = 1$, we get

$$\psi_{1,2} = \frac{2}{L}\sin\frac{\pi x}{L}\sin\frac{2\pi y}{L} \qquad E_{1,2} = \frac{5h^2}{8mL^2} \tag{4.22}$$

$$\psi_{2,1} = \frac{2}{L}\sin\frac{2\pi x}{L}\sin\frac{\pi y}{L} \qquad E_{2,1} = \frac{5h^2}{8mL^2} \tag{4.23}$$

Here, we see that, although the wavefunctions are different, they are degenerate. It means that they correspond to the same energy. In this case, there are two degenerate wavefunctions, and we say that the energy level is ‘doubly degenerate’. The occurrence of degeneracy is related to the symmetry of the system.